KHIN ANDREI GAMBOA

PASIG CITY, METRO MANILA | +63 992 421 5130

gamboa.khinandrei@gmail.com | github.com/sondre01 | linkedin.com/in/khinandreigamboa/

A fourth-year Computer Engineering student seeking an On-the-Job Training opportunity to gain hands-on experience in software development, embedded systems, AI, and IoT. Eager to learn, improve technical skills, and contribute to real-world engineering projects.

EDUCATIONAL BACKGROUND

Bachelor of Science in Computer Engineering
Rizal Technological University - Pasig Campus

TECHNICAL SKILLS

Programming Languages: C#, C/C++, Python, JavaScript
Database: MySQL
Embedded Systems: Arduino, Raspberry Pi
AI & IoT: Basic AI object detection, Multilayer Perceptron (MLP) and IoT-based data transmission
Networking: Basic networking concepts, Peer-to-peer connections and Common network topologies
Tools: Microsoft Word, Excel, PowerPoint and Figma

ACADEMIC PROJECTS

Restaurant Management System
C#, Windows Forms, Database

- Developed a desktop-based system for managing orders, billing, and inventory

Computer Parts and Repair Shop Management System
C#, Windows Forms, Database

- Designed a system to manage computer parts inventory, track repairs, maintain customer records, and automate billing.

Arduino-Based Embedded Systems Activities
Arduino, Raspberry Pi, Sensors, C/C++, Python

- Developed embedded systems projects using Arduino and Raspberry Pi, integrating sensors like MAX30102, joystick-controlled and servo-calibrated systems, with web interfaces (HTML, CSS, JS) and basic object/face recognition on Raspberry Pi.

AI Kilobot (Embedded Systems Project)
Raspberry Pi, 24kg Load Cell, AI Object Detection

- Developed an AI-based embedded system using Raspberry Pi
- Integrated a load cell sensor for accurate weight measurement
- Implemented AI object detection to identify objects
- Combined sensor data and AI output for system decision-making

THESIS PROJECT

Four-in-One Vital Sign Sensor with BMI Calculation using AI and IoT for Health Risk Prediction

Raspberry Pi, 24kg Load Cell, AI Object Detection

- Developed a kiosk-based health monitoring system measuring heart rate, respiratory rate, oxygen saturation (MAX30102), body temperature (MLX90614), and blood pressure using AI-based object detection on a digital BP monitor
- Measured weight and height using load cell (HX711) and TF Luna LiDAR for automatic BMI calculation
- Implemented AI using a multilayer perceptron (MLP) for data analysis, including object detection for lifestyle and wearable monitoring
- Enabled real-time data transmission via IoT, focusing on accuracy, reliability, and user-friendly health summaries